

Change Theme

Language

Java 15



Problem

Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m. Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq \text{array elements} \leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Submissions

Leaderboard

Discussions

```

1  import java.io.*;
2  import java.util.Scanner;
3  class Solution {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          int n = sc.nextInt();
7          int m = sc.nextInt();
8          int[] arr = new int[n];
9          for(int i=0;i<n;i++){
10             arr[i] = sc.nextInt();
11         }
12         int[] arr1 = new int[m];
13         for(int i=0;i<m;i++){
14             arr1[i] = sc.nextInt();
15         }
16         int[] merged = new int[n+m];
17         int i=0;
18         int j=0;
19         for(int k=0;k<n+m;k++){
20             if(i<n && (j>=m || arr[i]<=arr1[j]))
21                 merged[k]=arr[i];
22                 i++;
23             }else{
24                 merged[k]=arr1[j];
25                 j++;
26             }
27         }
28         int total = n+m;

```

Line: 38 Col: 1